
Impact Statement

Zimbabwe Agricultural Extension RAG System

The Zimbabwe Agricultural Extension RAG System is designed to give smallholder farmers fast, reliable access to expert agricultural advice on crop diseases, pests, nutrient deficiencies, soil management, and climate adaptation. By retrieving locally validated, expert-reviewed content rather than generating free-form text, the system aims to reduce farmers' dependence on scarce in-person AGRITEX extension officers, particularly in remote areas where visits may be infrequent. The most direct positive impact is timelier decision-making: a farmer facing an unfamiliar crop disease could receive relevant, vetted guidance in minutes rather than waiting weeks for an extension visit, potentially reducing crop losses and improving household food security.

However, the system also carries risks that unfold over time. Over-reliance on a digital tool could gradually erode informal knowledge-sharing networks between farmers and local extension officers, weakening social ties that have historically supported resilience during shocks such as drought or pest outbreaks. There is also a risk of unequal benefit: because the dataset is predominantly English-language and sourced from formally documented literature, Shona- and Ndebele-speaking farmers, and those in areas without smartphone or internet access, may see fewer benefits than better-resourced farmers, potentially widening existing agricultural inequalities rather than closing them.

On values, the system is intended to support farmer autonomy by presenting ranked source documents rather than prescriptive commands, allowing farmers to weigh advice against their own judgment and local conditions. It aims to uphold dignity by treating local and indigenous agricultural knowledge as equally valid to formal scientific literature, rather than positioning external expertise as inherently superior. Community wellbeing is supported insofar as improved yields and reduced crop loss contribute to household and village-level food security, though this benefit depends heavily on the accuracy and contextual relevance of the underlying documents.

Key uncertainties remain. It is not yet clear how retrieval quality will hold up for hyper-local conditions not well represented in the source literature, such as microclimate-specific pest behaviour, nor how farmers with lower digital literacy will interact with the tool in practice. To manage these risks, the team plans to maintain feedback channels with AGRITEX and farmer cooperatives, monitor for uneven usage patterns across language and geography, and continue to expand local-language and community-sourced content over time.

References

[1] Centre for the Governance of AI. (2020). A Guide to Writing the NeurIPS Impact Statement. Retrieved from <https://medium.com/@GovAI/a-guide-to-writing-the-neurips-impact-statement-4293b723832>